

PROJECT REPORT

Marketing Attribution & ROI Dashboard

An Interactive Business Intelligence Project on
Marketing Channel Performance, Attribution, and Return on Investment

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Table of Contents

1. Title Page	1
2. Abstract	3
3. Introduction	3
4. Problem Statement.....	3
5. Project Objectives.....	4
6. Technologies Used	4
7. Dataset Description	4
Ad Spend Data	4
CRM / Conversion Data.....	4
Web Analytics Data.....	4
8. Data Cleaning and Preprocessing	5
9. Data Modeling.....	5
Fact Table.....	5
Dimension Tables.....	5
10. KPI and DAX Calculations	6
11. Dashboard Development	6
11.1 Executive Dashboard	6
11.2 Marketing Performance Analysis	7
12. Key Insights	7
13. Challenges Faced	7
14. Conclusion.....	8
15. Future Enhancements	8

2. Abstract

Organizations today invest heavily across multiple marketing channels and campaigns, making it essential to understand which activities truly drive revenue and deliver the best return on investment. This project presents a Marketing Attribution and ROI Dashboard designed to analyze marketing performance, measure return on advertising spend, and support data-driven decision-making.

The project uses advertising spend data, CRM and conversion data, and web analytics data to build a unified view of marketing performance. Python and Pandas were used for data cleaning and preprocessing, SQL was used for data storage and querying, and Power BI with DAX was used to build an interactive dashboard for reporting key metrics.

Key performance indicators such as Total Revenue, Total Marketing Spend, Return on Ad Spend (ROAS), Conversion Rate, and Customer Acquisition Cost (CAC) were calculated to evaluate marketing effectiveness. The resulting dashboard allows stakeholders to compare channels, campaigns, products, and countries, and to identify which marketing efforts generate the highest value for the business.

This report documents the complete project workflow, from data preparation and modeling to KPI calculation and dashboard development, and presents the key insights and business value derived from the analysis.

3. Introduction

Marketing analytics plays a crucial role in helping organizations understand how their marketing investments translate into business outcomes. As businesses expand their presence across multiple channels, campaigns, and markets, it becomes increasingly important to measure performance accurately and consistently.

Dashboards provide a centralized, visual way for stakeholders to monitor performance without needing to manually analyze raw data. They allow marketing teams, analysts, and decision-makers to quickly identify trends, compare performance across dimensions, and respond to changes in a timely manner.

Measuring revenue and marketing performance together is essential to understanding the true impact of marketing activities. Simply tracking spend or leads in isolation does not reveal whether marketing efforts are generating meaningful business value.

Marketing attribution refers to the process of identifying which marketing channels, campaigns, or touchpoints contributed to a conversion or sale. Return on Investment (ROI), often measured through Return on Ad Spend (ROAS), evaluates how efficiently marketing spend is being converted into revenue. Together, attribution and ROI analysis form the foundation of this project.

4. Problem Statement

Organizations invest in multiple marketing channels and campaigns, but it can be difficult to identify which activities generate the highest revenue and provide the best return on investment. Without a consolidated view

of marketing performance, decision-makers may struggle to allocate budgets effectively or identify underperforming channels and campaigns.

This project addresses this problem by creating an interactive dashboard for analyzing marketing performance, enabling stakeholders to evaluate channel effectiveness, track conversions, and make informed, data-driven marketing decisions.

5. Project Objectives

The main objectives of this project are:

- Analyze marketing campaign performance.
- Measure revenue and marketing expenditure.
- Evaluate marketing effectiveness using ROAS.
- Analyze leads and conversion status.
- Compare channels, products, campaigns, and countries.
- Build an interactive dashboard for data-driven decision-making.

6. Technologies Used

Technology	Purpose
Python	Data cleaning and preprocessing
Pandas	Data manipulation
SQLite / SQL	Data storage and querying
Power BI	Dashboard development
DAX	KPI calculations

7. Dataset Description

The project uses three primary datasets, each contributing a different perspective on marketing and sales performance.

Ad Spend Data

Contains information related to:

- Campaign
- Marketing channel
- Advertising spend
- Date

CRM / Conversion Data

Contains:

- Lead or customer information
- Product
- Country
- Deal value
- Conversion status
- Conversion date

Web Analytics Data

Contains:

- Website activity
- Marketing touchpoints
- Channel information
- User interactions

8. Data Cleaning and Preprocessing

The following process was followed to prepare the raw data for analysis:

1. Loaded the datasets using Python.
2. Checked for missing values.
3. Identified duplicate records.
4. Removed unnecessary duplicates.
5. Converted incorrect data types.
6. Prepared the cleaned datasets for analysis.
7. Exported the processed data for SQL and Power BI.

Representative code snippets used during the cleaning process can be included here, for example the steps used to check for missing values, remove duplicates, and correct data types using Pandas.

9. Data Modeling

The data model for this project follows a star schema design, which separates measurable business data from descriptive attributes to support efficient querying and analysis in Power BI.

Fact Table

Fact_Marketing

Contains measurable data such as:

- Revenue
- Attributed Revenue

- Deal Value
- Marketing-related metrics

Dimension Tables

- Dim_Product
- Dim_Channel
- Dim_Country

These dimension tables provide descriptive information used to filter and analyze the fact data.

[Insert Power BI Model View Screenshot Here]

Figure 1: Power BI Data Model (Star Schema)

10. KPI and DAX Calculations

The following key measures were defined to evaluate marketing performance:

Total Revenue

Total Revenue = SUM(Revenue)

Total Marketing Spend

Total Spend = SUM(Ad Spend)

ROAS (Return on Ad Spend)

ROAS = Total Revenue ÷ Total Marketing Spend

Conversion Rate

Conversion Rate = (Won Deals ÷ Total Leads) × 100

CAC (Customer Acquisition Cost)

CAC = Total Marketing Spend ÷ Total Customers

The actual DAX formulas used to implement these measures in Power BI should be included in this section, along with brief notes on any filter context or calculated columns used to support them.

11. Dashboard Development

11.1 Executive Dashboard

This dashboard provides a high-level overview of overall marketing and revenue performance. It includes:

- KPI Cards
- Revenue by Marketing Channel
- Revenue by Product
- Revenue by Country
- Revenue Distribution
- Interactive Slicers

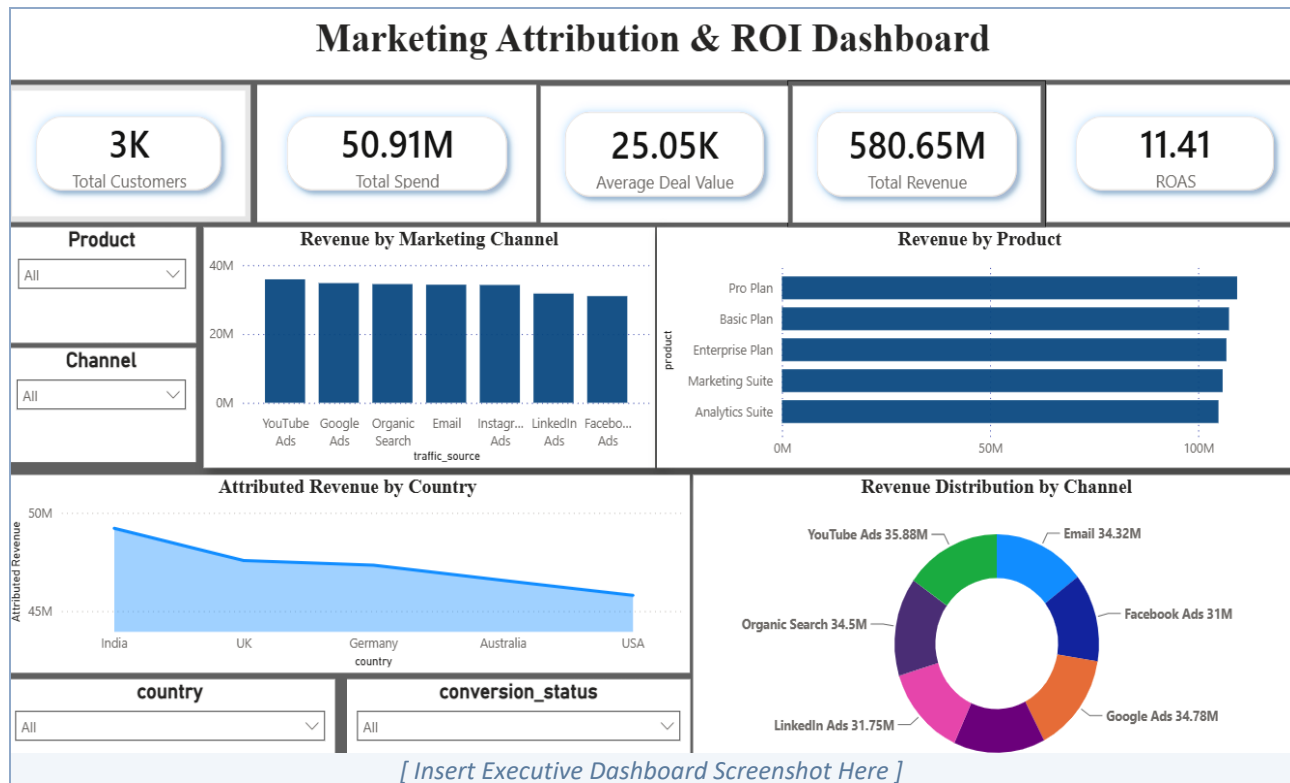


Figure 2: Executive Dashboard

11.2 Marketing Performance Analysis

This dashboard focuses on the effectiveness and efficiency of marketing activities. It includes:

- Marketing Spend by Campaign
- Leads by Country
- Marketing Spend Trend
- Monthly ROAS
- Conversion Status
- Leads Over Time

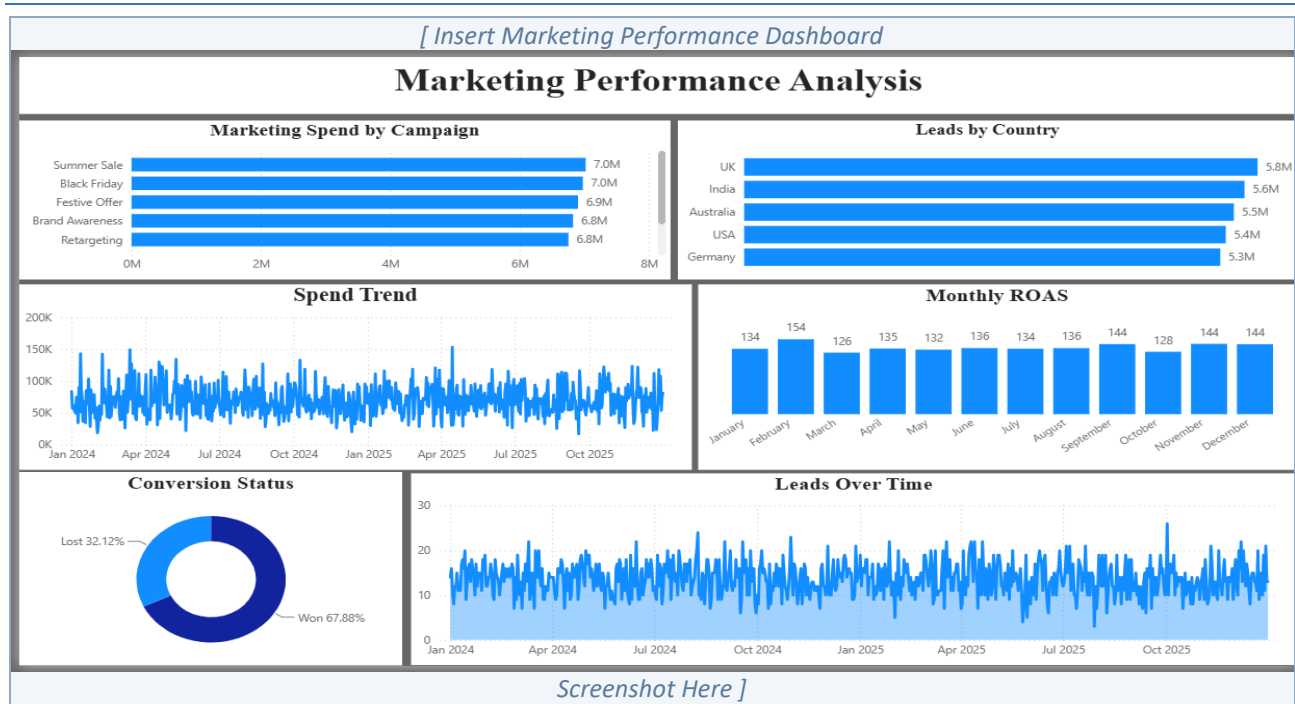


Figure 3: Marketing Performance Analysis Dashboard

12. Key Insights

The dashboard produced several important insights, including:

- Identification of high-performing marketing channels.
- Comparison of marketing spend between campaigns.
- Analysis of countries generating more leads.
- Monitoring of Won and Lost conversions.
- Analysis of changes in marketing spend over time.
- Evaluation of marketing efficiency using ROAS.

13. Challenges Faced

During the project, the following challenges were encountered:

- Handling missing and duplicate data.
- Creating relationships between different datasets.
- Managing different date fields.
- Creating correct DAX measures.
- Ensuring slicers filtered all relevant visuals.
- Designing a clear and professional dashboard.

14. Conclusion

This project successfully transformed raw marketing data from multiple sources into a single, interactive dashboard that supports informed business decision-making.

The dashboard helps businesses:

- Monitor performance
- Compare marketing channels
- Analyze revenue
- Evaluate ROAS
- Understand lead conversion
- Support data-driven decision-making

15. Future Enhancements

The project can be extended in the future through the following enhancements:

- Revenue forecasting
- Customer Acquisition Cost analysis
- Cost Per Lead analysis
- Cost Per Click analysis
- AI-based marketing recommendations
- Campaign anomaly detection
- Predictive analytics